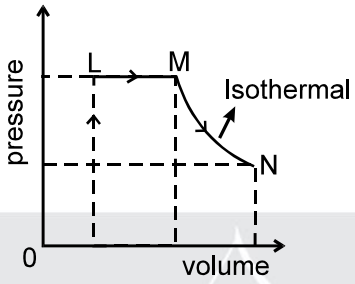
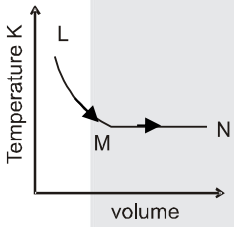
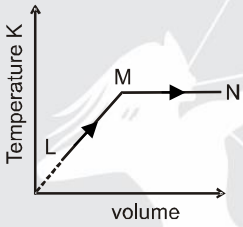
SYLLABUS : KTG & THERMODYNAMICS

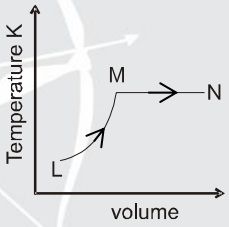
1. A fixed mass of an ideal gas undergoes changes of pressure and volume starting at L, as shown in Figure.

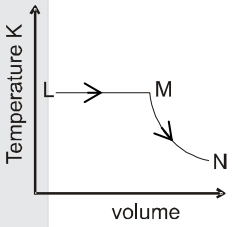


Which of the following is correct :

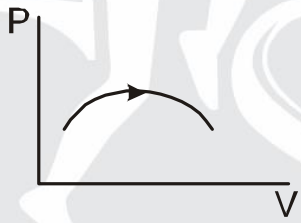
(A) 

(B) 

(C) 

(D) 

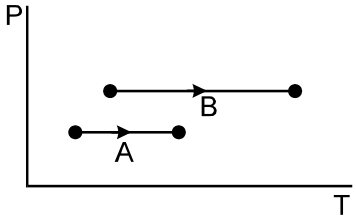
2. In figure, P-V curve of an ideal gas is given. During the process, the cumulative work done by the gas



(A) continuously increases (B) continuously decreases

(C) first increases then decreases (D) first decreases then increases

3. In given figure, let ΔW_1 and ΔW_2 be the work done by the gas in process A and B respectively then (given change in volume is same in both process)



(A) $\Delta W_1 > \Delta W_2$

(B) $\Delta W_1 = \Delta W_2$

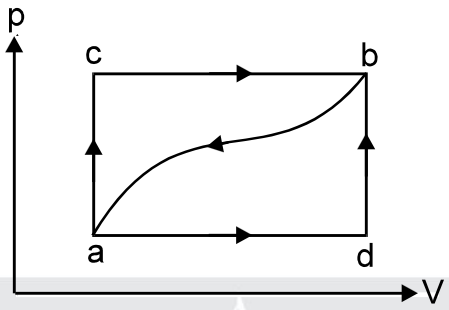
(C) $\Delta W_1 < \Delta W_2$

(D) Nothing can be said about the relation between ΔW_1 and ΔW_2

4. The internal energy of a gas is given by U = 5 + 2PV. It expands from V_0 to $2V_0$ against a constant pressure P_0. The heat absorbed by the gas in the process is :–

(A) $-3P_0V_0$ (B) $3P_0V_0$ (C) $2P_0V_0$ (D) P_0V_0

5. When a system is taken from state 'a' to state 'b' along the path 'acb', it is found that a quantity of heat Q = 200 J is absorbed by the system and a work W = 80J is done by it. Along the path 'adb', Q = 144J. The work done along the path 'adb' is



(A) 6J (B) 12 J (C) 18 J (D) 24 J

6. In the above question, if the work done on the system along the curved path 'ba' is 52J, heat abosrbed is

(A) – 140 J (B) – 172 J (C) 140 J (D) 172 J

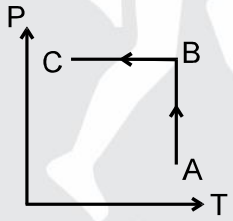
7. In above question, if U_a = 40J, value of $U_b$will be

(A) – 50 J (B) 100 J (C) – 120 J (D) 160 J

8. In above question, if U_d = 88 J, heat absorbed for the path 'db' is

(A) – 72 J (B) 72 J (C) 144 J (D) – 144 J

9. Ideal gas is taken through process shown in figure:



(A) In process AB, work done by system is positive

(B) In process AB, heat is rejected out of the system.

(C) In process AB, internal energy increases

(D) In process AB internal energy decreases and in process BC internal energy increases.

10. The value of the ratio C_p/C_v for hydrogen is 1.67 at 30 K but decreases to 1.4 at 300 K as more degrees of freedom become active. During this rise in temperature (assume H_2 as ideal gas),

(A) C_p remains constant but C_v increases

(B) C_p decreases but C_v increases

(C) Both C_p and C_v decrease by the same amount

(D) Both C_p and C_v increase by the same amount

11. Boiling water is changing into steam. Under this condition, the specific heat of water is

(A) zero (B) one (C) Infinite (D) less than one

12. For an ideal gas, the heat capacity at constant pressure is larger than than that at constant volume because

(A) positive work is done during expansion of the gas by the external pressure

(B) positive work is done during expasion by the gas against external pressure

(C) positive work is done during expansion by the gas against intermolecular forces of attraction

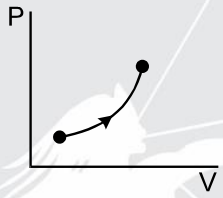
(D) more collisions occur per unit time when volume is kept constant.

13. A gas has :

(A) one specific heat only (B) two specific heats only

(C) infinite number of specific heats (D) no specific heat

14. If molar heat capacity of the given process (as shown in figure) is C, then



(A) $C < C_V$ (B) $C = 0$ (C) $C > C_v$ (D) $C = C_v$

15. For small positive coefficient of expansion in case of solid,

(A) $C_p - C_v = R$ (B) $C_p - C_v = 2R$

(C) C_p is slightly greater than C_v (D) C_p is slightly less than C_v

16. The molar specific heat of the process $V \propto T^4$ for CH_4 gas at room temperature is:–

(A) 4R (B) 7R (C) 3R (D) 8R

17. A gas is contained in a metallic cylinder fitted with a piston. The gas is suddenly compressed by pushing piston downward and is maintained at this position. After this process, as time passes the pressure of the gas in the cylinder

(A) increases

(B) decreases

(C) remains constant

(D) increases or decreases depending on the nature of the gas.

18. Two samples 1 and 2 are initially kept in the same state. The sample 1 is expanded through an isothermal process where as sample 2 through an adiabatic process upto the same final volume. The final temperature in process 1 and 2 are T_1 and T_2 respectively, then

(A) $T_1 > T_2$

(B) $T_1 = T_2$

(C) $T_1 < T_2$

(D) The relation between T_1 and T_2 cannot be deduced.

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19. Let P_1 and P_2 be the final pressure of the samples 1 and 2 respectively in the previous question then :

(A) $P_1 < P_2$ (B) $P_1 = P_2$ (C) $P_1 > P_2$

(D) The relation between P_1 and P_2 cannot be deduced.

20. Let ΔW_1 and ΔW_2 be the work done by the systems 1 and 2 respectively in the previous question then:

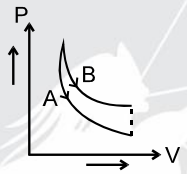
(A) $\Delta W_1 > \Delta W_2$ (B) $\Delta W_1 = \Delta W_2$ (C) $\Delta W_1 < \Delta W_2$

(D) The relation between W_1 and W_2 cannot be deduced.

21. A given quantity of a gas is at pressure P and absolute temperature T. The isothermal bulk modulus of the gas is:

(A) $\frac{2}{3}P$ (B) P (C) $\frac{3}{2}P$ (D) 2P

22. In figure, A and B are two adiabatic curves for two different gases. Then A and B corresponds to :



(A) Ar and He respectively (B) He and H_2 respectively

(C) O_2 and H_2 respectively (D) H_2 and He respectively

23. The internal energy of a gas in an adiabatic process is given by U = a + bPV, find γ :–

(A) $\frac{a+1}{a}$ (B) $\frac{b+1}{b}$ (C) $\frac{b+1}{a}$ (D) $\frac{a}{b+1}$

24. A gas undergoes a process in which its pressure P and volume V are related as VP^n = constant. The bulk modulus of the gas in the process is :

(A) nP (B) $P^{1/n}$ (C) P/n (D) P^n

25. The work of 146 kJ is performed in order to compress one kilo mole of a gas adiabatically and in this process the temperature of the gas increases by 7 ^oC. The gas is (R = 8.3 J mol^{-1} K^{-1})

(A) diatomic (C) triatomic

(C) mixture of monoatomic and diatomic (D) monoatomic

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ANSWER KEY

1.	(B)	**2.**	(A)	**3.**	(C)	**4.**	(B)	**5.**	(D)
6.	(B)	**7.**	(D)	**8.**	(B)	**9.**	(B)	**10.**	(D)
11.	(C)	**12.**	(B)	**13.**	(C)	**14.**	(C)	**15.**	(C)
16.	(B)	**17.**	(B)	**18.**	(A)	**19.**	(C)	**20.**	(A)
21.	(B)	**22.**	(B)	**23.**	(B)	**24.**	(C)	**25.**	(A)